# Finiteness of Shallow Logical Operators and a Sharp Depth Threshold

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For a quantum code on $n$ finite-dimensional physical subsystems, we consider all exact logical unitary circuits of depth at most $h \geq 1$, with gates on at most $\rho$ subsystems. For $\rho \geq 2$ and code distance $d \geq 2$, these circuits have only finitely many logical actions, up to phase, whenever $h < 2\lceil \log_\rho d \rceil - 1$, where no stabilizer or geometric locality assumption is needed, and a gauge subsystem is allowed. In contrast, for every qubit stabilizer code encoding at least one qubit, an infinite logical family is available at depth $2\lceil \log_\rho d \rceil - 1$, for $\rho \geq 2$ and $d \geq 2$. This is the exact critical depth for the onset of infinitely many logical actions in that class.

During a quantum computation, circuit depth controls the spread of physical errors. The gate range $\rho$ is the maximum number of physical subsystems on which an elementary gate may act. A single fault in a depth-$h$ circuit can affect at most $\rho^h$ subsystems, so constant-depth circuits of bounded range limit error propagation between error-correction steps. Controlling this spread is central to fault-tolerant computation. Under its code and decoding assumptions, Gottesman's threshold construction protects polynomial-size computations below a noise threshold with constant space overhead [1]. We therefore ask which logical operations can be realized at shallow depth and how much depth is needed for universal logical control.

*[To be revised.]* Early Clifford-hierarchy bounds impose geometric or algebraic assumptions beyond gate range and distance. Bravyi and König considered local stabilizer checks and geometrically local gates [2]. Extensions to Pauli subsystem codes and geometrically nonlocal gates use further cleaning or disjointness conditions [3, 4]. Finiteness also appears for protected gates in two-dimensional topological field theories [5], where equivalence can remove separate phases on invariant sectors. Here only one global phase is removed. Webster et al. restrict universal gate sets for infinite-disjointness stabilizer-code families [6], an assumption not imposed here.

Under one common total-depth bound, universal approximation requires infinitely many exact logical actions. We therefore ask at what depth this necessary condition can first hold. For $\rho \geq 2$ and $d \geq 2$, Theorem 1 rules out an infinite logical image when $h < 2\lceil \log_\rho d \rceil - 1$, without a stabilizer or geometric locality assumption. The finiteness criterion does not assign a Clifford hierarchy level. Theorem 2 shows that every qubit stabilizer subspace code encoding at least one qubit realizes a nontrivial continuous logical family at depth $2\lceil \log_\rho d \rceil - 1$, when arbitrary gates on at most $\rho$ qubits are allowed. Stabilizer subspace codes are thus the most capable base models for the onset of an infinite logical family, attaining the smallest possible depth at given $d$ and $\rho$. Figure 1 illustrates the two mechanisms.

Let the physical system be $\mathcal{H} = \bigotimes_{i=1}^{n} \mathcal{H}_i$, with $n \geq 1$ and finite-dimensional factors. The code $\mathcal{C}$ is specified by an isometry $V : \mathcal{H}_L \otimes \mathcal{H}_G \longrightarrow \mathcal{H}$, where $\dim \mathcal{H}_L = K \geq 2$ and $\dim \mathcal{H}_G = g \geq 1$. The factors carry logical and gauge degrees of freedom, respectively. A distance lower bound $d$ means that every operator $O$ supported on fewer than $d$ physical subsystems satisfies

$$V^\dagger O V = I_L \otimes M_O \tag{1}$$

for some operator $M_O$ on $\mathcal{H}_G$ [7, 8]. A physical unitary $U$ implements an exact logical action when

$$UV = V(W \otimes Z) \tag{2}$$

for unitaries $W$ on $\mathcal{H}_L$ and $Z$ on $\mathcal{H}_G$. This equality holds for all logical and gauge states, and the logical action is the phase class $[W] \in \mathrm{PU}(K)$. For integers $h \geq 1$ and $\rho \geq 2$, let $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ be the set of these classes obtained when $U$ has at most $h$ layers of arbitrary unitary gates on disjoint supports of at most $\rho$ physical subsystems. The depth bound applies to each complete circuit.

*[To be revised.]* Eastin–Knill restricts transversal gates through a compact Lie-group argument [9]. Circuits subject to a common depth bound need not form a group, and successive layers can have intersecting supports. Our proof applies the infinitesimal restriction to circuit parameters using both forward and backward light cones. Detectable sums of generators on overlapping erasable regions have also been studied in the theory of covariant error correction [10, 11]. Polynomial circuit maps, their conjugated gate-insertion derivatives, and semialgebraic stratification are established tools in quantum circuit analysis [12]. The required code condition is error detection, allowing gauge degrees of freedom [7, 8]. Our proof uses a finite smooth decomposition of a polynomial constraint set [13].

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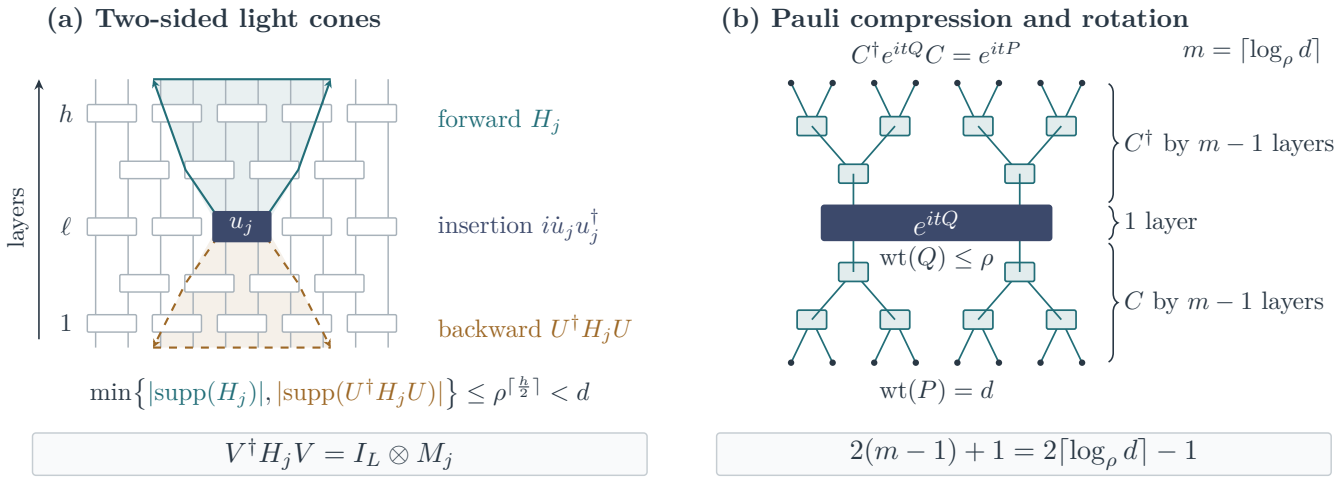



FIG. 1. Mechanisms behind Theorems 1 and 2. Time runs upward in both panels. (a) An insertion at gate $u_j$ in layer $\ell$ has forward and backward supports bounded by $\rho^{h-\ell+1}$ and $\rho^\ell$. When $\rho^{\lceil \frac{h}{2} \rceil} < d$, at least one support is detectable. The exact encoded action transfers its gauge-only compression to $V^\dagger H_j V = I_L \otimes M_j$, for a gauge operator $M_j$. The brickwall arrangement is schematic and imposes no geometric locality. (b) With $m = \lceil \log_\rho d \rceil$ and $\rho \ge 2$, Clifford gates compress a weight-$d$ logical Pauli to $Q = CPC^\dagger$ in $m-1$ layers. One gate $e^{itQ}$ and $m-1$ inverse layers give $e^{itP}$ at depth $2m-1$. The case $(\rho, d) = (2, 8)$ is illustrated. In general, a gate may collect up to $\rho$ factors. Only active Pauli support is shown. The physical qubits remain.

**Theorem 1.** *Let $n \ge 1$, $h \ge 1$, and $\rho \ge 2$ be integers, and let $V$ encode a quantum code $\mathcal{C}$ satisfying Eq. (1) for an integer $d \ge 2$ on $n$ physical subsystems. If*

$$h < 2\lceil \log_\rho d \rceil - 1, \tag{3}$$

*then the logical actions of all circuits of depth at most $h$, with gates on at most $\rho$ subsystems, that satisfy Eq. (2) obey*

$$|\mathcal{L}_{\le h}(\mathcal{C}, \rho)| < \infty.$$

*All gate supports and unitary matrices are allowed. Consequently, $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$ is not dense in* $\mathrm{PU}(K)$.

*Proof.* Equation (3) is equivalent to $\rho^{\lceil \frac{h}{2} \rceil} < d$.

Group all circuits by the gate supports in each layer. After adding singleton identity gates on idle sites, a layer is a partition of the $n$ sites into blocks of size at most $\rho$. There are finitely many such partitions, and hence finitely many support arrangements for $h$ layers. Circuits of smaller depth are included by identity padding. The unitary matrices on these supports remain arbitrary.

Choose one arrangement, denoted by $\alpha$, and list its gates in layer order as $u_1, \ldots, u_J$, so that $U = u_J \cdots u_1$. Consider a smooth family of gate matrices and logical factors $W, Z$ satisfying Eq. (2). Its Hermitian generator is

$$H = i\dot{U}U^\dagger = \sum_{j=1}^{J} U_{>j}(i\dot{u}_j u_j^\dagger)U_{>j}^\dagger, \tag{4}$$

where $U_{>j} = u_J \cdots u_{j+1}$. This is the conjugated gate-insertion derivative used in circuit parameter analysis [12].

Write the $j$th summand as $H_j$, and suppose that gate $j$ is in layer $\ell$. Its forward light cone has at most $\rho^{h-\ell+1}$ sites. In the opposite frame,

$$U^\dagger H_j U = U_{<j}^\dagger (i u_j^\dagger \dot{u}_j) U_{<j},$$

where $U_{<j} = u_{j-1} \cdots u_1$. This operator has a backward light cone of at most $\rho^\ell$ sites. Other gates in layer $\ell$ have disjoint supports and cancel from both counts. At least one of these two supports therefore has size at most

$$\rho^{\min\{\ell, h-\ell+1\}} \le \rho^{\lceil \frac{h}{2} \rceil} < d.$$

If the forward support is smaller than $d$, Eq. (1) applies directly to $H_j$. If only the backward support is smaller, it gives $V^\dagger U^\dagger H_j U V = I_L \otimes N_j$. Equation (2) then implies

$$\begin{aligned} V^\dagger H_j V &= (W \otimes Z)(I_L \otimes N_j)(W^\dagger \otimes Z^\dagger) \\ &= I_L \otimes Z N_j Z^\dagger. \end{aligned}$$

Thus $V^\dagger H_j V = I_L \otimes M_j$ for every $j$, with all compressions taken in the same frame. Summing gives

$$V^\dagger H V = I_L \otimes M \tag{5}$$

where $M = \sum_{j=1}^{J} M_j$ is Hermitian on $\mathcal{H}_G$. This is the detectable-generator step [9, 10].

Differentiating Eq. (2), and using $U^\dagger V = V(W^\dagger \otimes Z^\dagger)$, also gives

$$\begin{aligned} V^\dagger H V &= A \otimes I_G + I_L \otimes B, \qquad (6) \\ A = i\dot{W}W^\dagger, &\qquad B = i\dot{Z}Z^\dagger. \end{aligned}$$

Taking the partial trace over $\mathcal{H}_G$ yields

$$A = \frac{\mathrm{Tr}(M - B)}{g}\, I_L.$$

Thus $\dot{W} = -i\lambda W$ for a real scalar $\lambda$, and the logical motion is only a phase.

To obtain finiteness, let $D_j$ be the Hilbert-space dimension on the support of gate $j$, and put

$$\mathcal{P}_\alpha = \left(\prod_{j=1}^{J} \mathrm{U}(D_j)\right) \times \mathrm{U}(K) \times \mathrm{U}(g).$$

Let $\mathcal{X}_\alpha \subseteq \mathcal{P}_\alpha$ consist of the tuples $(\mathbf{u}, W, Z)$ satisfying Eq. (2). Unitarity and Eq. (2) are polynomial equations in the real and imaginary matrix entries, so $\mathcal{X}_\alpha$ is a real algebraic set.

Lemma 1 in the End Matter gives a finite smooth decomposition

$$\mathcal{X}_\alpha = \bigsqcup_{r=1}^{N_\alpha} C_r, \qquad N_\alpha < \infty, \tag{7}$$

where each cell is smoothly diffeomorphic to an open cube $(0,1)^{\nu_r}$, including a point when $\nu_r = 0$. This partition also covers singular points of $\mathcal{X}_\alpha$.

For two tuples $x, y \in C_r$, choose a smooth diffeomorphism $\varphi_r : (0,1)^{\nu_r} \to C_r$, and set $a = \varphi_r^{-1}(x)$, $b = \varphi_r^{-1}(y)$. The path

$$\gamma(t) = \varphi_r\big((1-t)a + tb\big), \qquad 0 \le t \le 1, \tag{8}$$

stays in $C_r$. Its coordinates give smooth gate matrices and logical factors satisfying Eq. (2) throughout. The preceding calculation makes $[W]$ constant along $\gamma$. Each cell therefore contributes at most one logical operation modulo phase, giving at most $N_\alpha$ operations for this support arrangement.

The finitely many arrangements cover every circuit of depth at most $h$ and gate range at most $\rho$. Thus $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$ is a finite union of finite sets. Since $K \ge 2$, it is closed and cannot be dense in $\mathrm{PU}(K)$. □

Finiteness does not exclude a finite subset $S \subseteq \mathcal{L}_{\le h}(\mathcal{C}, \rho)$ for which

$$\overline{\langle S \rangle} = \mathrm{PU}(K).$$

The group $\langle S \rangle$ allows words of arbitrary finite length. A word of length $r$ in $S \cup S^{-1}$ has a concatenated implementation of depth at most $rh$. These products are not required to remain in $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$, so density of the generated group does not assert density within the original depth bound. For example, Dua et al. give a $[[37, 1, 7]]$ CSS code with separate depth-one circuits of gate range $\rho = 3$ that implement [14, Sec. V and Appendix B 3]

$$S = \{\overline{T}^\dagger, \sqrt{\overline{X}}^{\,\dagger}\} \subseteq \mathcal{L}_{\le 1}(\mathcal{C}, 3).$$

The two circuits use different partitions of the physical qubits. Since $\overline{H} \propto (\overline{T}^\dagger)^2 \sqrt{\overline{X}}^{\,\dagger} (\overline{T}^\dagger)^2$ and $\overline{T} = (\overline{T}^\dagger)^7$, this pair generates a dense subgroup of PU(2). For this code, $2\lceil \log_3 7 \rceil - 1 = 3$, so $\mathcal{L}_{\le 1}(\mathcal{C}, 3)$ is finite by Theorem 1.

In a fault-tolerance threshold construction based on the constant-depth unitary logical gadgets considered here, logical-gate compilation is necessary for universal control in the finite-image regime. On a fixed code, the available shallow logical actions form a finite alphabet. If its generated group is dense, arbitrary target logical unitaries can be approximated by words over this alphabet, with error correction interleaved as prescribed by the protocol [1, 14]. The constant-depth premise applies to each gadget. Compiled word lengths cannot remain uniformly bounded when arbitrary targets must be approximated to arbitrarily high accuracy.

The finiteness condition has an exact boundary for stabilizer codes. For an integer $\rho \ge 2$, define the critical depth

$$h_\infty(\mathcal{C}, \rho) = \min\{h \in \mathbb{Z}_{\ge 1} : \mathcal{L}_{\le h}(\mathcal{C}, \rho) \text{ is infinite}\}, \tag{9}$$

with value $\infty$ if the set is empty. For $d \ge 2$, this definition and Theorem 1 give

$$h_\infty(\mathcal{C}, \rho) \ge 2\lceil \log_\rho d \rceil - 1. \tag{10}$$

The matching upper bound uses established Pauli-rotation synthesis. Binary-tree compression appears in Ref. [15], Sec. IV and Fig. 2(a). Cowtan et al. give balanced-tree phase gadgets with logarithmic CNOT depth [16, Cor. 3.4 and the following discussion], and Haah, Liu, and Tan explicitly obtain logarithmic-depth Pauli rotations with all-to-all connectivity [17, Cor. 3.6]. Choosing a weight-$d$ logical Pauli gives the logarithmic upper bound for stabilizer codes as a direct application. The depth-one case with gate range at least $d$ is also noted explicitly by Dua et al. [14, Sec. I], using a minimum-weight logical Pauli as a rotation generator.

**Theorem 2.** *Let $\mathcal{C}$ be an $[[n, k, d]]$ qubit stabilizer code, with $k \ge 1$ and actual distance $d \ge 2$. Let $\rho \ge 2$ be an integer, and allow arbitrary unitary gates on any set of at most $\rho$ qubits. Then*

$$h_\infty(\mathcal{C}, \rho) = 2\lceil \log_\rho d \rceil - 1. \tag{11}$$

*At this depth, the logical image contains a nontrivial continuous one-parameter subgroup.*

*Proof.* The lower bound is Eq. (10). For a stabilizer code, the distance $d$ is the minimum weight of a Hermitian $n$-qubit Pauli operator that preserves the code space and acts nonscalarly on it. Choose such an operator $P$ of weight $d$, with $P^2 = I$, and put $m = \lceil \log_\rho d \rceil$.

Partition the nonidentity factors of $P$ into groups of $\rho$, except possibly one smaller group. On each group, basis changes and CNOTs conjugate the product to a Pauli on

one qubit. Their composition is a single allowed Clifford gate on that group. The groups are disjoint, so these gates form one layer. If $w_r$ is the weight after $r$ layers, then $w_{r+1} = \lceil \frac{w_r}{\rho} \rceil$ and $w_0 = d$, so $w_r = \lceil \frac{d}{\rho^r} \rceil$. Hence $m-1$ layers give a Clifford circuit $C$ such that

$$Q = CPC^\dagger, \qquad \mathrm{wt}(Q) \le \rho.$$

Local basis changes are included in the allowed Clifford gates and require no additional layers. When $m = 1$, take $C = I$. The family

$$U(t) = C^\dagger e^{itQ} C = e^{itP} \tag{12}$$

has depth at most $2m-1$, since its middle factor is a single allowed gate. It preserves the code and induces $e^{it\overline{P}}$, where $\overline{P}$ is a nontrivial logical Pauli. Since $\overline{P}$ has both eigenvalues $\pm 1$, these projective actions have period $\pi$ and form a nontrivial continuous U(1) subgroup. □

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## END MATTER

### Finite smooth decomposition of polynomial zero sets

Together with the infinitesimal argument of Theorem 1, finite Whitney stratification of the semialgebraic constraint set into connected smooth strata already gives a finite logical image. To keep the argument elementary and self-contained, we give a short proof using linear algebra and the implicit function theorem, with elementary polynomial arithmetic. The finite decomposition below is the polynomial case of standard semialgebraic theory [13]. A smooth cell below means an embedded smooth submanifold that is smoothly diffeomorphic to an open cube $(0,1)^r$, with a point allowed when $r = 0$. No smoothness of the full zero set is assumed.

**Lemma 1.** *For any finite family of real polynomials on $\mathbb{R}^N$, there is a finite partition of $\mathbb{R}^N$ into smooth cells on each of which every polynomial has constant sign. Consequently, the common zero set of finitely many real polynomials is a finite disjoint union of smooth cells.*

*Proof.* We induct on $N$. In one variable, the finitely many real roots of the nonzero polynomials partition the line into points and open intervals. The assertion is also immediate for $N = 0$. Write the variables in the induction step as $(x,t)$, where $x \in \mathbb{R}^{N-1}$. Discard identically zero input polynomials, and write the remaining ones as $p_1(x,t), \dots, p_a(x,t)$.

We first specify a finite family of projection polynomials in $x$. For every nonempty subset $I \subseteq \{1,\dots,a\}$, form $F_I = \prod_{i\in I} p_i$ and include all its coefficients in $t$. For every truncation

$$f(x,t) = \sum_{j=0}^{m} f_j(x) t^j$$

of $F_I$, with $1 \le m \le \deg_t F_I$, also include all square minors of the coefficient matrix of the linear map

$$\begin{aligned} T_f : (A,B) &\longmapsto Af + B\partial_t f, \\ \deg A \le m-1, &\qquad \deg B \le m. \end{aligned} \tag{13}$$

The target consists of polynomials of degree at most $2m-1$, so this matrix has $2m$ rows and $2m+1$ columns. Its entries and minors are polynomials in $x$. Zero polynomials may be discarded. There are finitely many subsets, truncations, coefficients, and minors. By induction, partition the base $\mathbb{R}^{N-1}$ into finitely many smooth cells on which these projection polynomials have constant signs.

Fix one base cell $C$. Each $p_i$ is either identically zero as a polynomial in $t$ at every $x \in C$, or has one constant degree there. This follows from the coefficient signs. If all are identically zero, the cylinder $C \times \mathbb{R}$ is already a suitable cell. Otherwise let $I$ be the indices of the nonzero factors and set $F = F_I$. Its degree $m$ is constant on $C$. If $m = 0$, the same cylinder suffices. Suppose $m \geq 1$. The truncation at this degree agrees with $F$ on every fiber over $C$, so its projected matrix represents $T_F$.

The projected minors make the rank of $T_F$ constant on $C$. At a point $x$, write $F = d_0 f_0$, $\partial_t F = d_0 g_0$, where $d_0$ is the monic greatest common divisor and $f_0, g_0$ are coprime. Put $s = \deg d_0$ and $r = m - s$. Then $\deg f_0 = r$, $\deg g_0 = r - 1$, and $r \geq 1$. The identity $AF + B\partial_t F = 0$ holds exactly when $A = Qg_0$ and $B = -Qf_0$. The degree restrictions in Eq. (13) give $\deg Q \leq s$. Hence

$$\dim \ker T_F = s + 1, \qquad \operatorname{rank} T_F = 2m - s.$$

In particular, $s$ and $r$ are constant on $C$.

The monic squarefree part $G = \frac{f_0}{c}$, where $c$ is the leading coefficient of $F$, has precisely the real roots of $F$, each simple: differentiation lowers each root multiplicity by one. We show that its coefficients depend smoothly on $x \in C$, even where $F$ has repeated roots. Solve

$$\begin{aligned} AF + B\partial_t F = 0, \qquad & \deg A < r, \\ B = t^r + \sum_{j<r} b_j t^j. \qquad & \end{aligned} \tag{14}$$

There is exactly one solution: $A = -\frac{g_0}{c}$, $B = \frac{f_0}{c}$. The coefficient matrix for the $2r$ unknown coefficients is a column submatrix of $T_F$. It has full column rank: a homogeneous solution would have $\deg B < r$, whereas any nonzero kernel vector has $B = -Qf_0$ of degree at least $r$. Choose a nonzero $2r$-by-$2r$ minor of this coefficient matrix at one point of $C$. It is among the projected minors and thus remains nonzero throughout $C$. Cramer's rule expresses every coefficient in Eq. (14) as a rational function of $x$ with a denominator that does not vanish on $C$. Thus $G = B$, the monic squarefree part, has smooth coefficients on $C$. The same construction for the singleton products gives a smooth monic squarefree part $G_i$ for every nonconstant active factor $p_i$.

The number of real roots of $G$ is locally constant on $C$. Indeed, at a point $x_0$, its real roots are simple. Apply the implicit function theorem in a smooth coordinate chart of $C$: each root persists uniquely and smoothly. A monic polynomial $t^r + \sum_{j<r} b_j t^j$ has all its real roots in $|t| \leq 1 + \max_j |b_j|$. Near $x_0$, this gives one uniform bound. Outside small disjoint intervals about the old roots, $G(x_0, t)$ is bounded away from zero on the remaining compact interval. Continuity excludes new roots there. Within the small intervals, the derivative stays nonzero and there is exactly one root. Since $C$ is connected, the number of real roots is constant, and ordering them gives globally smooth functions

$$\theta_1(x) < \cdots < \theta_b(x).$$

Inactive factors vanish identically, and active constant factors have constant nonzero signs. For an active nonconstant factor, suppose $p_i(x_0, \theta_j(x_0)) = 0$. The simple root of $G_i$ at that point persists as a smooth root $\beta(x)$. Because $p_i$ divides $F$, this is also a root of $G$, and the local uniqueness of the latter root gives $\beta(x) = \theta_j(x)$. Thus vanishing on the graph is an open condition in $C$. Nonvanishing is open by continuity. Connectedness makes one alternative hold everywhere, and any nonzero sign is then constant.

Partition $C \times \mathbb{R}$ into these $b$ root graphs and the $b + 1$ intervening open bands, including the unbounded bands. No active factor vanishes in a band, so its sign is constant there. A graph is diffeomorphic to $C$. A band is diffeomorphic to $C \times (0, 1)$. For finite endpoints use affine interpolation. For a half-line use $\frac{u}{1-u}$, and for the whole line use $\frac{u}{1-u} - \frac{1-u}{u}$, where $0 < u < 1$. Thus all pieces are smooth cells. There are finitely many base cells, and each yields finitely many pieces. This completes the induction. The common zero set is the union of those cells on which all the given polynomials vanish. □

This construction includes nullified factors, degree drops, repeated roots, and singular points of the zero set. The subset products handle factors that vanish on a base cell. The truncations handle degree drops. The projected minors separate rank changes. Nothing requires the polynomial zero set itself to be a manifold.

To apply the lemma, express the unitary and encoding constraints for one circuit-support arrangement in real and imaginary matrix entries. Their common zero set is $\mathcal{X}_\alpha$. For two points in one cell, a smooth chart from an open cube carries the straight segment between their preimages to a smooth path in $\mathcal{X}_\alpha$. The circuit calculation in Theorem 1 gives $\dot{W} = -i\lambda W$ along this path, so its endpoints have the same projective logical action. Each cell contributes at most one such action. Finitely many cells, followed by finitely many support arrangements, give the claimed finiteness.